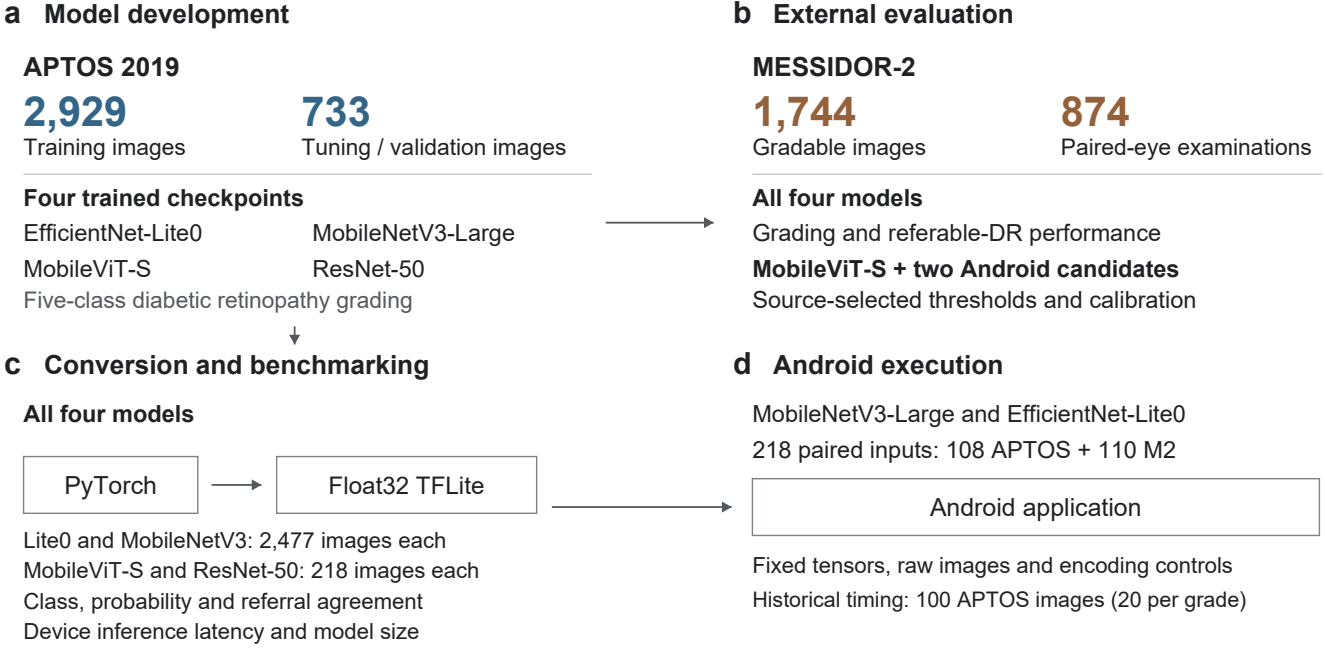

a Model development
APTOS 2019
2,929
Training images
733
Tuning / validation images
Four trained checkpoints
EfficientNet-Lite0
MobileNetV3-Large
MobileViT-S
ResNet-50
Five-class diabetic retinopathy grading
b External evaluation
MESSIDOR-2
1,744
Gradable images
874
Paired-eye examinations
All four models
Grading and referable-DR performance
MobileViT-S + two Android candidates
Source-selected thresholds and calibration
c Conversion and benchmarking
All four models
PyTorch
Float32 TFLite
Lite0 and MobileNetV3: 2,477 images each
MobileViT-S and ResNet-50: 218 images each
Class, probability and referral agreement
Device inference latency and model size
d Android execution
MobileNetV3-Large and EfficientNet-Lite0
218 paired inputs: 108 APTOS + 110 M2
Android application
Fixed tensors, raw images and encoding controls
Historical timing: 100 APTOS images (20 per grade)